DEVREL ENGINEER EXERCISE · RUNPOD

GPU Pre-flight

Check an open-source AI repo before you spin up a GPU for it.

GitHub API | OSV.dev API | FastAPI on a RunPod CPU Pod

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Before you pay for GPU time, four questions

find a repo → clone it onto a GPU pod → pip install → run it, billed by the hour

Is anyone maintaining it?

Bus factor, contributors, last push, archived

GitHub

Am I allowed to use it?

License

GitHub

Will the install match?

Share of dependencies pinned to exact versions

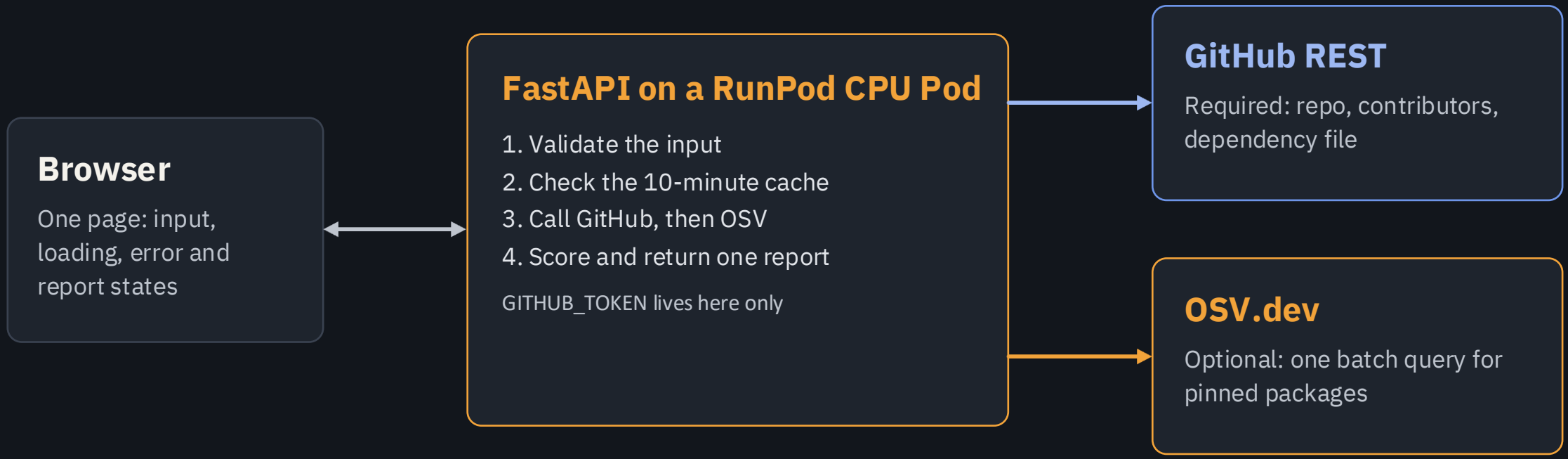
GitHub

Is what it installs safe?

Known vulnerabilities per pinned package

OSV.dev

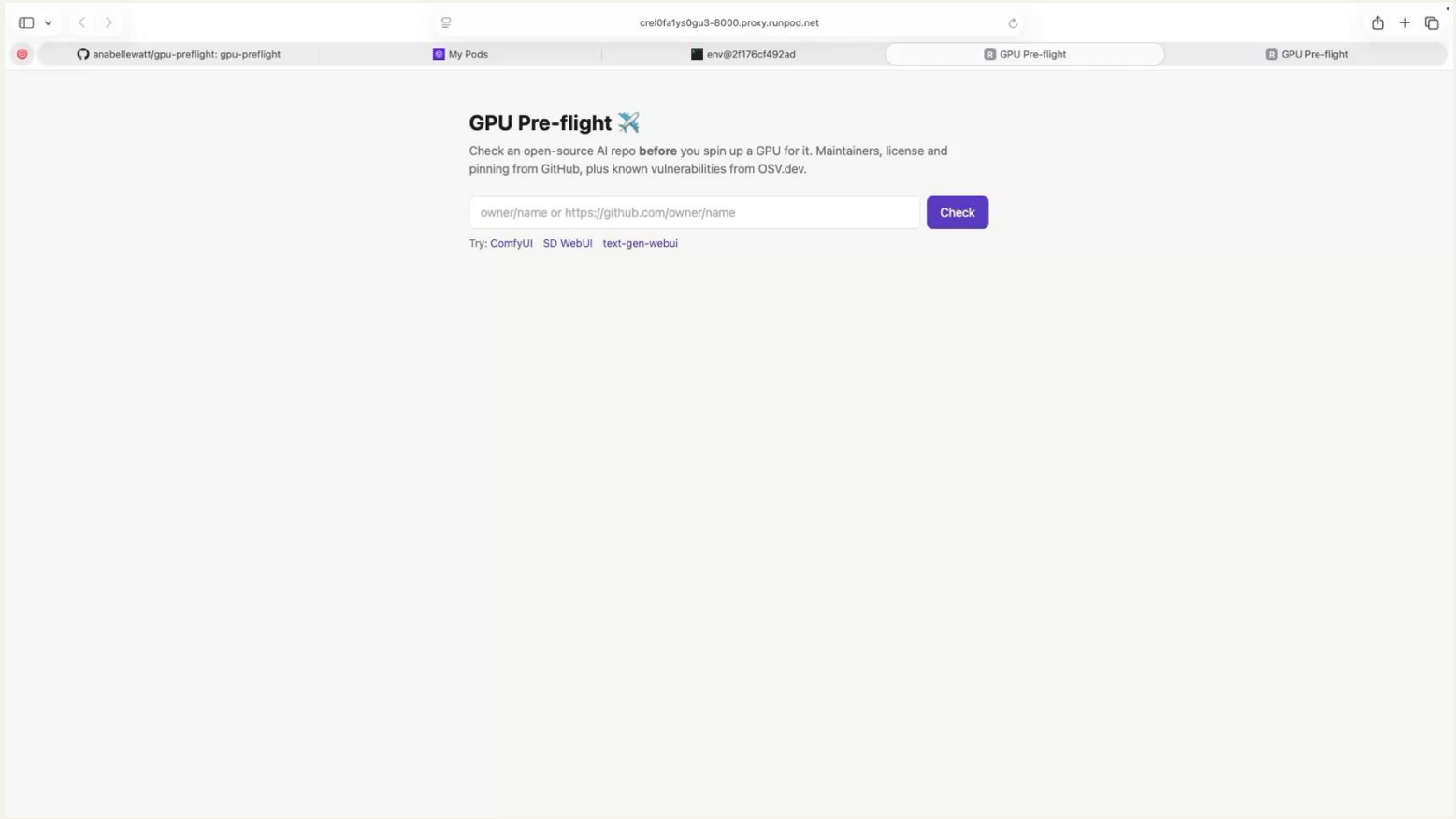
One deploy, one URL, two APIs



The same FastAPI app serves the page, so the browser never sees the token or calls an API directly.

DEMO 1 OF 3

Normal case: both APIs answer



WATCH FOR

1 · Three GitHub calls

Repo, contributors, requirements.txt

2 · One OSV.dev batch

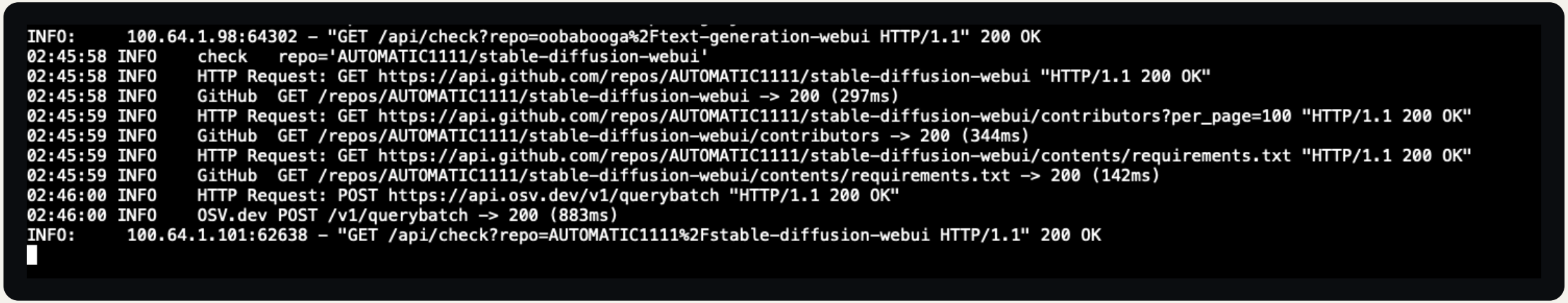
Every pinned package in one request

3 · One combined report

Score, bus factor, license, vulnerable packages

crel0fa1ys0gu3-8000.proxy.runpod.net

One click, traced through both APIs



1 · GitHub · 297ms

Repo info: license, last push, archived

2 · GitHub · 344ms

Contributors, for the bus factor

3 · GitHub · 142ms

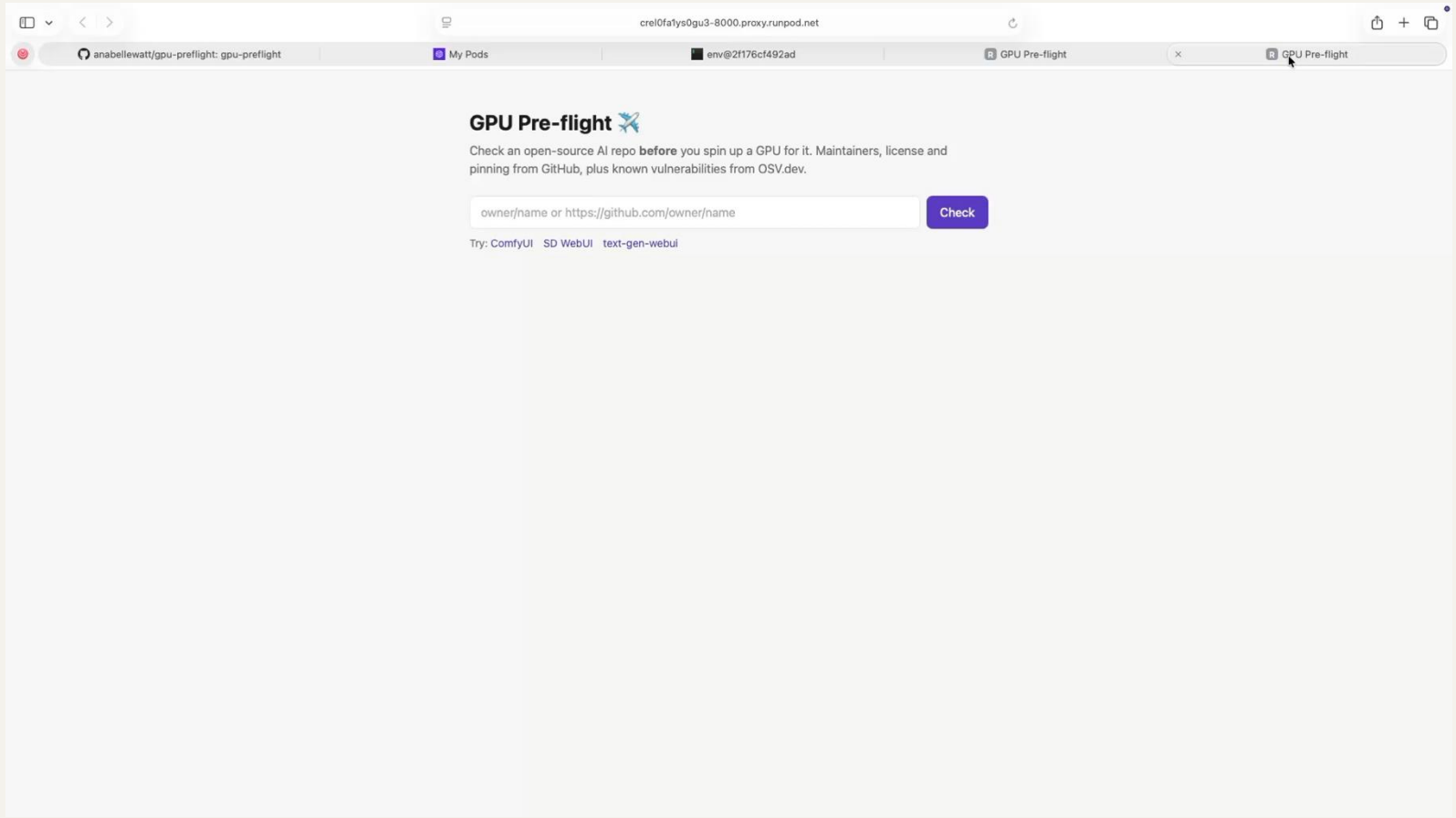
requirements.txt: what it installs

4 · OSV.dev · 883ms

One batch query: are the pinned packages vulnerable?

DEMO 2 OF 3

OSV.dev goes down: the report still loads



WATCH FOR

1 · Yellow banner and score higher

Vulnerabilities marked "not checked"

2 · GitHub data still there

Maintainers, license, last push

3 · "simulated outage" in the log

Works only when the pod runs with DEMO_MODE=1

crel0fa1ys0gu3-8000.proxy.runpod.net/?simulate=osv_down

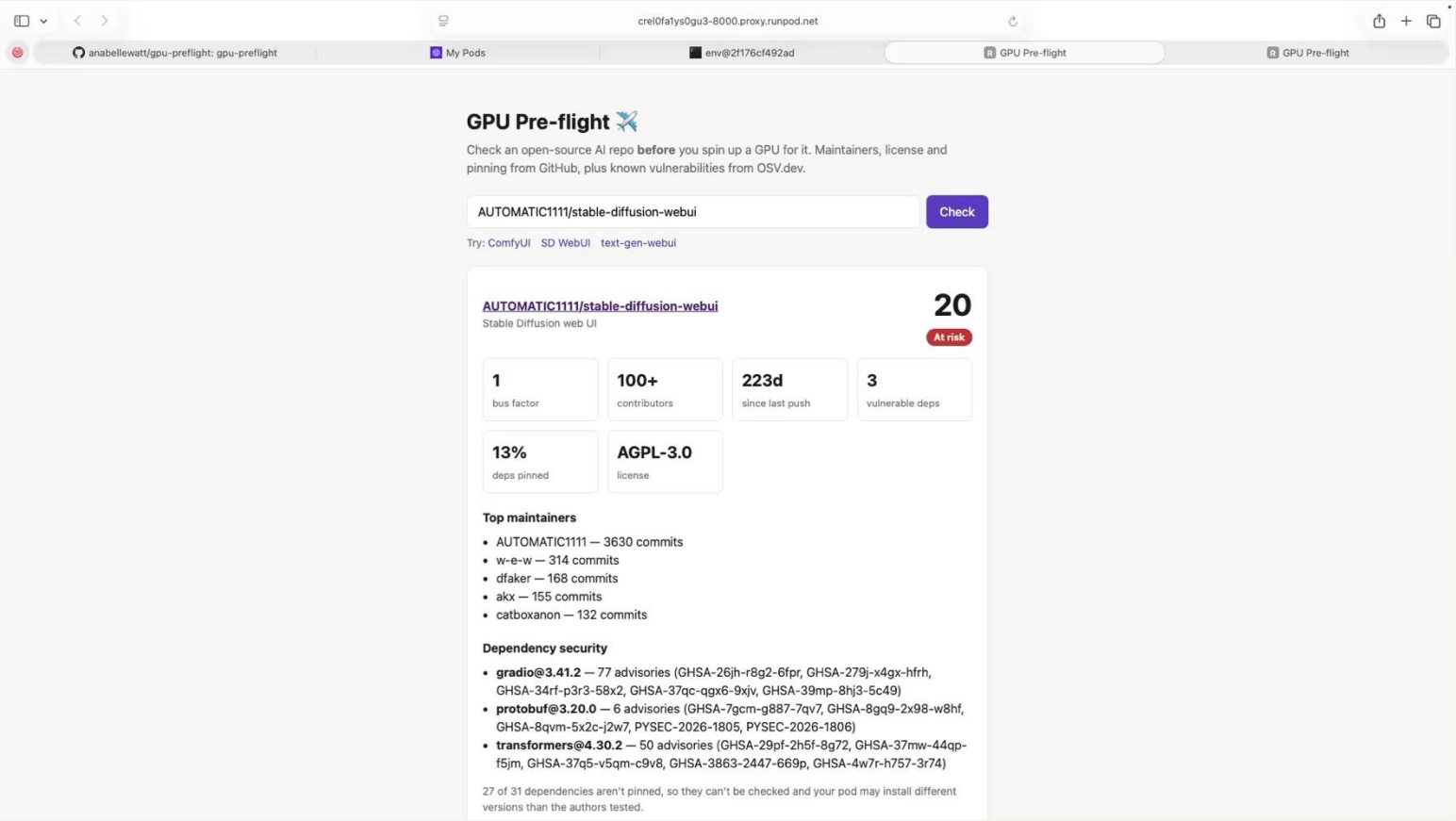
When an API fails, the user still knows what happened

Situation	What the user sees	HTTP
Invalid input	Message, and no upstream call is made	400
Repo not found	"Repository not found (or it is private)"	404
GitHub rate limit	"Try again in about N min", never retried	429
GitHub timeout or 5xx	Retried once, then a clear error	504 / 502
OSV.dev down	Report still loads, with a warning banner	200
Repo renamed	Redirect followed, report shows the new name	200

Live drill: crel0fa1ys0gu3-8000.proxy.runpod.net/?simulate=osv_down

DEMO 3 OF 3

Same repo again: served from the cache



WATCH FOR

1 · First check: full trace

GitHub and OSV calls in the log

2 · Second check: no API calls

Only the check line, and an instant answer

3 · Why it matters

10-minute cache protects the GitHub rate limit

The backend on its own: a JSON API



```
anabellewatt@Mac gpu-preflight % curl -s "https://crel0fa1ys0gu3-8000.proxy.runpod.net/api/check?repo=not-a-repo"
{"error":"Enter a GitHub repo as owner/name or a github.com URL"}%
anabellewatt@Mac gpu-preflight % curl -s "https://crel0fa1ys0gu3-8000.proxy.runpod.net/api/check?repo=AUTOMATIC1111/stable-diffusion-webui" | python3 -m json.tool | head -3
0
{
    "repo": "AUTOMATIC1111/stable-diffusion-webui",
    "url": "https://github.com/AUTOMATIC1111/stable-diffusion-webui",
    "description": "Stable Diffusion web UI",
    "stars": 165039,
    "days_since_push": 223,
    "license": "AGPL-3.0",
    "open_issues": 2510,
    "archived": false,
    "maintainers": {
        "bus_factor": 1,
        "contributors": 100,
        "top": [
            {
                "login": "AUTOMATIC1111",
                "commits": 3630
            },
            {
                "login": "w-e-w",
                "commits": 314
            },
            {
                "login": "dfaker",
                "commits": 168
            },
            {
                "login": "akx",
                "commits": 155
            },
            {
```

Bad input stops early

400 with a clear message, before any API call

Same JSON the page renders

Usable from a script or CI, not just the browser

Bus factor 1

165k stars, yet one person made over half the commits among its top 100 contributors

The challenge: AI repos move

GET /repos/oobabooga/text-generation-webui → 301 Moved Permanently

GET /repositories/580642043 → 200 OK

Symptom

ComfyUI and text-generation-webui failed. Stable Diffusion WebUI worked.

Cause

Both repos were renamed. GitHub answers 301, and httpx doesn't follow redirects by default.

Fix

follow_redirects=True, plus a regression test for renamed repos.

Tradeoffs I made, and what comes next

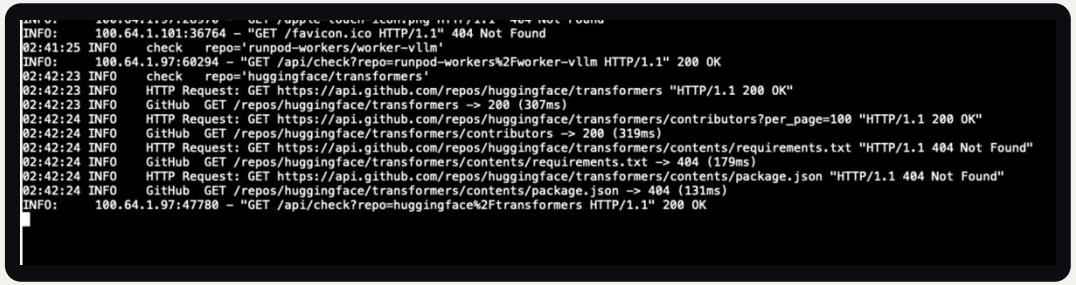
TRADEOFFS

Only pinned packages go to OSV

OSV needs an exact version. Unpinned ones are counted as a reproducibility risk instead of guessed at.

Only root requirements.txt or package.json

huggingface/transformers uses pyproject.toml, so OSV is skipped:



WITH MORE TIME

- Parse pyproject.toml and lockfiles
- Use the canonical name after a redirect: half the GitHub calls
- Match pinned CUDA and torch to RunPod images
- A "Deploy to RunPod" button for repos that pass
- Docker template so the pod starts the app on boot
- Shared cache (Redis) to scale across pods

How I used AI

Claude

- Proposed use cases and the RunPod framing
- Scaffolded the backend, frontend and tests
- Drafted the README and these slides
- Diagnosed the 301 from my log and wrote the fix

Me

- Chose the idea and the AI-repo angle
- Ran and tested it locally; found the renamed-repo failures
- Deployed it on a RunPod CPU pod
- Traced live requests and picked the demo repos
- Reviewed every file so I can explain and change it